

SOLVE-A-THON

TRACK 1 (BCA) - Problem Statement: Departmental Digital Resource & Knowledge Hub

Context & Background

Currently, academic resources (notes, question banks, textbooks) are scattered across WhatsApp groups, personal emails, and physical photocopies. This fragmentation leads to students missing crucial study materials. Furthermore, there is no verified "central truth" for official notices, leading to misinformation.

The objective is to build a centralized **Web Portal**—similar to the **DHI** or College ERP interface—that serves as a digital library and notice board for the department. It must facilitate a structured flow of information between the **HOD (Admin)**, **Faculty**, and **Students**.

User Roles (The Actors)

The system must support three distinct types of users, each with different permissions:

1. **The Admin (HOD):** The Super User who oversees the entire system.
2. **Faculty (Teachers):** The Content Creators and Approvers.
3. **Students:** The Consumers and Contributors (Peer-to-Peer).

Module Breakdown (What to Build)

1. Authentication & Security Module

- **Student Registration:** Students must sign up using their USN (University Seat Number), Name, and Semester.
- **Login:** Secure login for all three roles.
- **Validation:** Prevent duplicate accounts (e.g., One USN cannot have two accounts).

2. The "Smart Repository" (Upload & Download)

This is the core feature. Materials must be organized logically (Semester, Subject, Unit).

- **Supported Formats:** The system must accept uploads in exactly **4 formats**:
 1. **PDF** (Documents/Notes)
 2. **PPT/PPTX** (Presentations)
 3. **DOC/DOCX** (Assignments)
 4. **JPG/PNG** (Snapshots of whiteboard/handwritten notes)

- **Faculty Uploads:** Teachers can upload **Textbooks** (large files), **Reference Links** (URLs), and **Official Notes**. These bypass approval and go live immediately.
- **Student Uploads:** Students can upload their own notes, but these **must not go live immediately**. They enter a "Pending Approval" state.

3. The "Gatekeeper" System (Admin Approval Logic)

- **The Problem:** Students might upload wrong files or spam.
- **The Solution:**
 - When a student uploads a file, it is flagged as Status: Pending.
 - The **HOD/Faculty** dashboard has a "Review Uploads" tab.
 - They can **View** the file and click **Approve** (Publish to everyone) or **Reject** (Delete).
 - Only Status: Approved files are visible in the search results.

4. Advanced Search & Filtering Engine

Students should not have to scroll endlessly. They must be able to filter resources using a dropdown mechanism:

- **Filter by Subject Code:** (e.g., "BCA401")
- **Filter by Semester:** (e.g., "4th Sem")
- **Filter by Professor:** (e.g., "Notes by Prof. Sharma")
- *Outcome:* A clean grid view of files matching the criteria.

5. Digital Notice Board (Dashboard Highlights)

- A dedicated section on the Homepage called "**Department Circulars.**"
- **HOD/Teachers** can post urgent notices.
- **Visual Cue:** New notices should have a "New" tag or be highlighted in a different color for 24 hours.

NOTE: Ideas to Gain Extra Points (Bonus Features)

Students are encouraged to implement advanced features to distinguish their project and gain bonus points from the judges